Project Initialization and Planning Phase

Date:	21 October 2026	**Team ID:**	Piyush-Balodiya-GenAI-2026
Project Title:	CONSTRUCTAI – Multimodal Construction Intelligence Platform	**Maximum Marks:**	5 Marks

Project Proposal (Proposed Solution) Report

The proposal report aims to transform construction site monitoring, worker safety, and structural quality audits using multimodal Generative AI. It eliminates human observational blind spots, automates IS 456 crack classification, and provides real-time risk mitigations for high-value construction infrastructure projects.

Category	Details & Strategic Implementation
Project Overview Objective	Revolutionize construction site safety auditing and structural defect inspection by deploying real-time multimodal vision models (Groq LLaMA 3.2 Vision), achieving instant PPE compliance tracking and automated civil crack severity diagnostics.
Project Overview Scope	An end-to-end multimodal intelligence platform that ingests CCTV and drone site imagery, audits OSHA/IS PPE safety gear, diagnoses structural concrete cracks against Indian Standard IS 456:2000, and automates material BOQ estimations alongside live weather risk alerts.
Problem Statement Description	Manual site rounds fail to continuously monitor vast, multi-acre construction zones, resulting in unflagged PPE non-compliance, catastrophic worker fall injuries, and delayed structural crack diagnostics that lead to multi-lakh budget overruns and severe project delays.
Problem Statement Impact	Eliminating these inefficiencies achieves zero statutory OSHA penalties, prevents fatal on-site casualties, accelerates engineering sign-offs by 80%, and provides actionable chemical remediation steps before structural concrete defects worsen.
Proposed Solution Approach	Deploying high-speed multimodal vision inference powered by Groq LLaMA 3.2 Vision alongside a high-performance asynchronous FastAPI backend, connected to a responsive React.js dashboard tailored for high-contrast outdoor daylight use.
Proposed Solution Key Features	• **Real-Time PPE Detection:** Automated identification of missing hard hats, high-visibility vests, and harnesses. • **IS 456 Structural Crack Diagnostics:** Instant classification into hairline, shrinkage, or shear cracks with chemical repair blueprints. • **Weather & BOQ Risk Advisory:** Doppler weather sync to prevent concrete pour washouts and automate quantity takeoffs. • **Automated Audit Reports:** One-click executive PDF compliance reports generated with zero external data retention.

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Resource Requirements & Environment Specifications

Detailed resource allocation across compute hardware, development frameworks, third-party vision APIs, and benchmark datasets required to deploy and run CONSTRUCTAI efficiently.

Resource Type	Description	Specification / Allocation (CONSTRUCTAI)
Hardware **Computing Resources**	CPU & GPU specifications for local development and edge inference.	Intel Core i5 (12+ Cores) / NVIDIA RTX Series GPU & Cloud LPUs via Groq
Hardware **Memory (RAM)**	System RAM for concurrent vision pipeline and ASGI workers.	16 GB DDR5 RAM (minimum 8 GB required)
Hardware **Storage**	Fast SSD disk space for site imagery, model weights, and logs.	512 GB NVMe SSD Storage
Software **Backend Framework**	Asynchronous server-side API framework.	FastAPI (Python 3.10+) with Uvicorn ASGI Server
Software **Frontend Framework**	Client-side user interface and visualization dashboard.	React.js (Vite), Tailwind CSS, Lucide React UI
Software **Core Libraries**	Computer vision, data handling, and report generation packages.	groq-python, Pydantic, Pillow, ReportLab, SQLite3, Requests
Dev Environment **IDE & Tooling**	Integrated development environment and execution scripts.	VS Code / Cursor, Git Version Control, Windows Batch (.bat)
Data & Models **Multimodal AI Vision**	Primary vision-language model for multimodal safety and crack audits.	Groq LLaMA 3.2 11B / 90B Vision Preview (Sub-second LPUs)
Data & Models **Datasets & Standards**	Benchmark datasets and civil engineering regulatory codes.	Roboflow PPE Safety Benchmark + Concrete Crack Diagnostic Dataset (IS 456:2000 Guidelines)